

EDUCATION

University of Southern California

Master of Science, Geographic Information Science and Technology

Los Angeles, CA

Expected Dec 2026

Los Andes University,

Bachelor of Architecture and Design

Bogota, Colombia

April, 2020

GIS PROJECTS

GeoAI-Lite Water Main Break Prediction Tool – Seattle Water Utility

(Python GIS Programming & Customization – SSCI 586)

- Built a Python/ArcPy tool that automates preprocessing, spatial feature creation, and dataset assembly for water-main break prediction.
- Engineered variables using DEM elevation, road proximity, pipe attributes, and spatial joins to prepare machine-learning-ready data.

Spatially Explicit Wildfire ABM – Marshall Fire, Boulder County, CO

(Spatial Analysis – SSCI 583)

- Built an agent-based wildfire spread and suppression model in NetLogo using real GIS inputs, including NLCD land cover, Boulder County roads, and the Marshall Fire burn perimeter.
- Processed and exported spatial data in ArcGIS Pro, assigned land-cover-based flammability values, modeled wind-driven spread and spotting behavior, and created firefighter agents to test suppression scenarios.

Colombia Amphibian Explorer – Interactive Biodiversity Web Map

(Cartography and Visualization – SSCI 589)

<https://cortesju.github.io/amphibian-explorer/>

- Developed a custom interactive web map using the ArcGIS JavaScript API and GitHub Pages to visualize iNaturalist observations for 24 endemic and threatened amphibian species across Colombia.
- Cleaned and enriched VGI observation data with species names, IUCN conservation status, photos, audio links, timestamps, and supporting environmental context layers, including land cover, climate, precipitation, and protected areas.

EXPERIENCE

Las Flores Water Company

GIS Intern research

Altadena, CA

Summer 2025

The primary task was to support post-fire recovery by collecting field data and building a FEMA-ready report to fund water-meter repairs after the 2025 Altadena (Eaton) Fire.

- Collected meter condition, photos, and GPS points using ArcGIS Survey123.
- Standardized attributes and verified data quality for analysis and mapping.
- Created a map and a concise report used in the FEMA Public Assistance funding request.

Story Map developed : <https://storymaps.arcgis.com/stories/85bff3144ac048faae7432be7e494d5c>

University of Southern California

Los Angeles, CA

Teacher Assistant Human Populations and Natural Hazards

August 2025 to December 2025

The primary responsibility was to support instruction and applied learning for an undergraduate course focused on the interaction between human populations, natural hazards, and environmental risk, with an emphasis on geospatial analysis. initiatives.

- Assisted with course instruction covering floods, droughts, wildfires, earthquakes, sea level rise, and climate-related hazards.
- Supported students in applying GIS and spatial data analysis to evaluate hazard exposure, vulnerability, and risk.
- Reviewed and provided feedback on assignments involving ArcGIS Online StoryMaps, spatial datasets, and hazard impact assessments.

WSP 1 Bogota Metro line

Bogota, Colombia

Design Architect

February 2021 to November 2022

The primary task was to create the architectural design for four metro stations along Bogotá's first metro line

- Collaborated with multidisciplinary teams to meet technical standards
- Developed detailed plans using AutoCAD
- Applied GIS to support spatial decisions in station placement
- impacts on water sustainability

Los Andes University

Bogota, Colombia

Research student assistant,

July 2019 to December 2019

Contributed to water management studies for Bogotá's river systems, focusing on sustainable urban planning and territorial analysis. Leveraged GIS technology and demographic data to support environmental assessment and urban development research.

- Collaborated with multidisciplinary teams to meet technical standards
- Developed detailed plans using AutoCAD
- Applied GIS to support spatial decisions in station placement
- impacts on water sustainability

ADDITIONAL INFORMATION

Applications: ArcGIS Pro, ArcPy, Python, R, RStudio, GeoPandas, Jupyter Notebooks, scikit-learn (basic), GitHub, QGIS, Revit, AutoCAD, Adobe Creative Suite, Rhinoceros, SketchUp, Lumion, Blender

Languages: Fluent in Spanish

Organizations: Member, American Association of Geographers (AAG), Member, Association of Pacific Coast Geographers (APCG)

Awards: Tom McKnight & Joan Clemens Award for Outstanding Paper by a Master's Student – Association of Pacific Coast Geographers (2025)

Best story map in 2025 ArcGis StoryMaps Competition – Esri